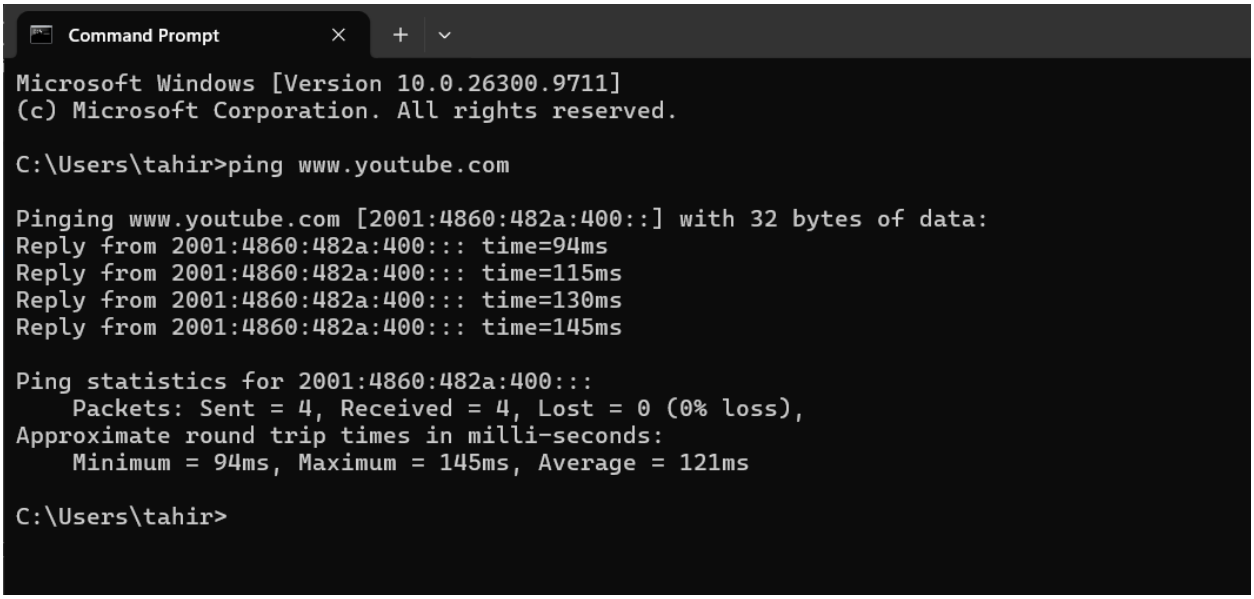
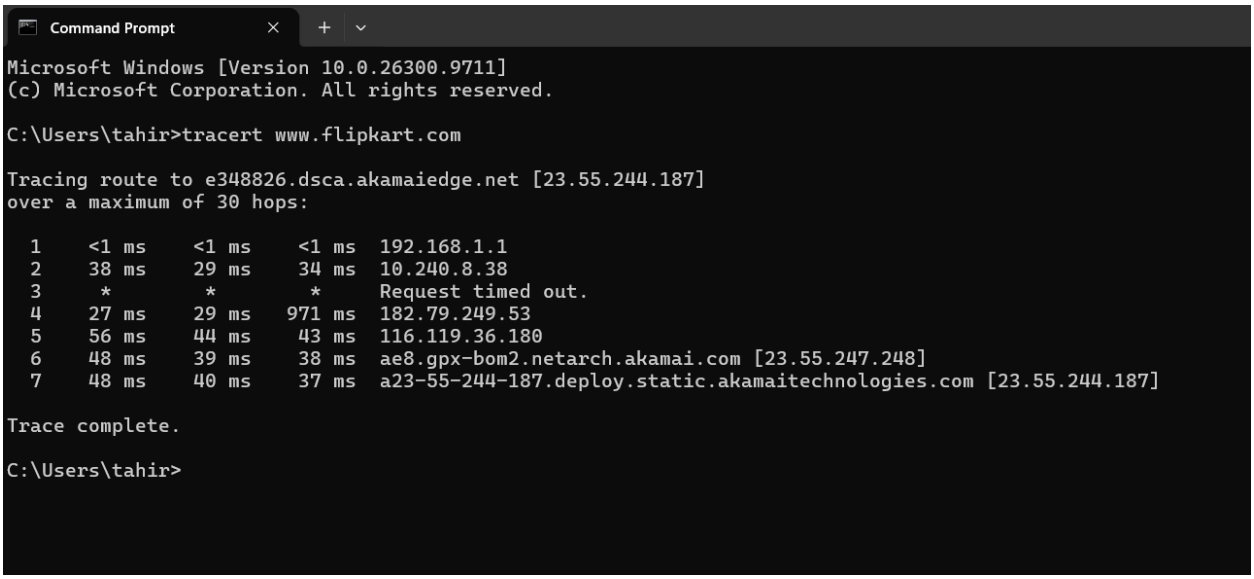
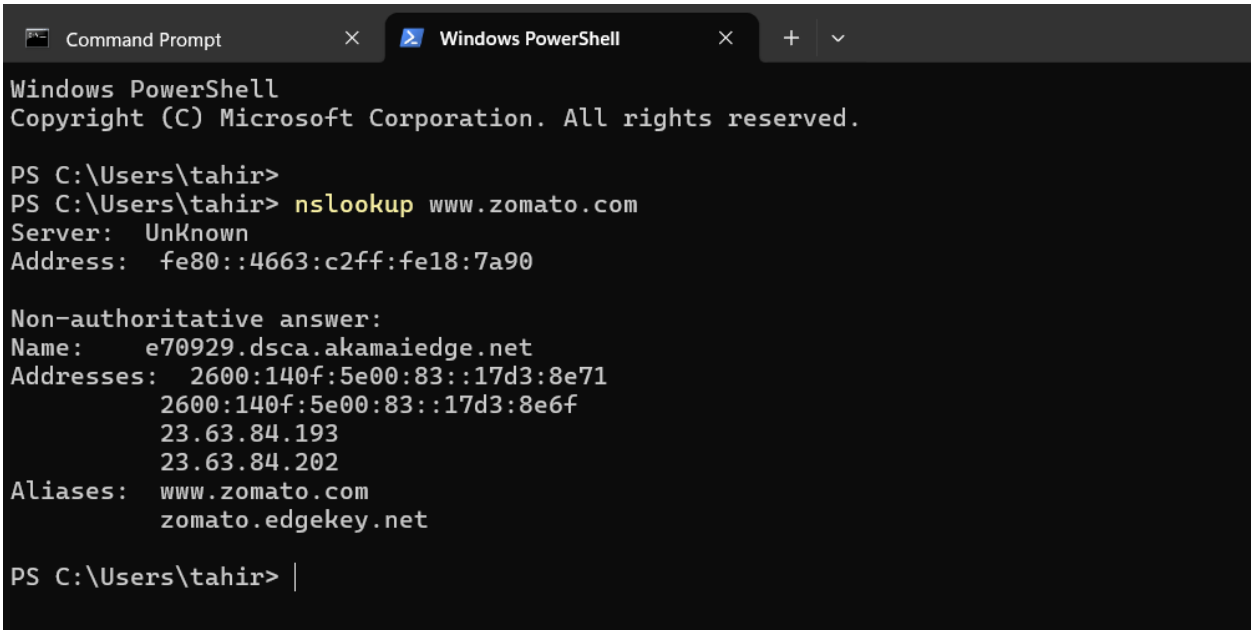
1. **Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.**



2. **Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output. Hint: Each hop represents a network device your data passes through to reach the destination.**

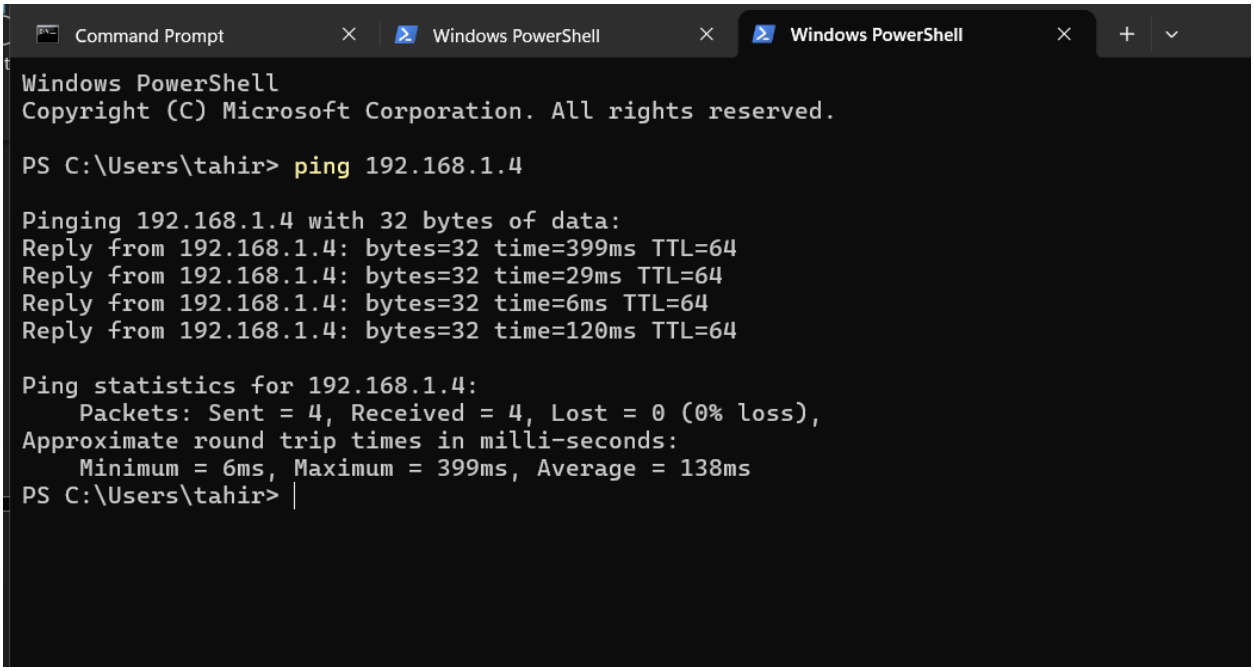


3. **Use the nslookup tool to find the IP address of www.zomato.com and record the result.**



4.

Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.



- **If the device is offline, the ping command will show "Request timed out" because the device is not responding to the ping request.**

5. **Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues. Constraint: Give a specific example for each tool based on your own test results above.**

1. **Website not loading**: I use ping to check if the website is responding.
2. **Slow connection**: I use traceroute to find where the connection is slow.
3. **DNS issues**: I use nslookup to check if a website name, like google.com, is getting the correct IP address.
4. When I ping an offline device, I may get “Request timed out.”

❖ These three tools help me find and understand network problems easily.